

Polymarket Market-Maker Bots for Crypto

15-Minute Binary Markets

Executive summary

Polymarket's "15-minute crypto" markets are short-horizon, two-outcome contracts (e.g., **Up** vs **Down**) that resolve by comparing the asset's price at the end of a 15-minute window to a "Price to Beat" defined at the start of that window; Polymarket's own market pages describe resolution as **Up if end price $\geq$ start price, else Down**, using **Chainlink data streams** as the resolution source (examples shown for ETH, SOL, XRP and analogous BTC markets). ¹

A market-making bot on these products is best modeled as an ultra-short-dated digital option market maker, but the most important practical driver of many bots' reported smooth (near-linear) equity curves is often **fee economics** rather than "alpha forecasting." Polymarket's trading fee model charges **taker-only, price-dependent fees** in certain categories (including Crypto), with fees computed as $\text{fee} = C \times \text{feeRate} \times p \times (1 - p)$ and **makers paying zero fees**; in Crypto the **taker fee rate is 0.072** and the **maker rebate share is 20%**. ²

Those taker fees fund a **daily USDC maker rebate program** where rebates are distributed proportionally to the **fee-equivalent** value of a maker's executed liquidity, using the same $C \times \text{feeRate} \times p \times (1 - p)$ weighting per fill. This creates a strong structural incentive to quote near the midpoint (where $p \times (1 - p)$ is maximal), because it maximizes "fee-equivalent" weight per executed share. In steady volume regimes, this can translate into a relatively steady rebate stream, which—when combined with spread capture—can resemble linear growth. ³

Technically, Polymarket is a **hybrid CLOB + on-chain settlement** system. Outcome positions are ERC-1155 tokens under the **Gnosis Conditional Token Framework (CTF)** on Polygon; tokens are fully collateralized by USDC.e (as documented today), created via **split**, recombined via **merge**, and redeemable after resolution. ⁴ Settlement occurs through Polymarket exchange contracts on Polygon (e.g., **CTF Exchange**), while order entry and matching involve off-chain components and authenticated APIs. ⁵

To build or evaluate a market-maker bot rigorously, you should (a) treat strategy P&L as a combination of **spread, rebates, adverse selection, tail risk near expiry**, and operational frictions (tick-size changes, latency, restarts, rate limits), and (b) backtest with data that can reproduce *fills* (not just midpoints). Polymarket provides rich endpoints for prices, orderbooks, trades, and account snapshots, but historical full depth-of-book is not guaranteed—so serious studies generally require **your own market-data capture pipeline** using the WebSocket market channel. ⁶

Polymarket 15-minute crypto market structure

Contract definition and lifecycle

Polymarket's dedicated "15 Min Crypto" collection shows continuously rolling short-duration markets across major assets (BTC, ETH, SOL, XRP and others), each representing a 15-minute window. ⁷ Individual market pages specify:

- **Outcome rule:** "Up" resolves if the asset's price at the end of the specified window is **greater than or equal to** its price at the beginning; otherwise "Down." ¹
- **Resolution source:** Chainlink price "data stream" is explicitly named (e.g., ETH/USD, with similar language for SOL and XRP). ⁸
- **Market dynamics:** because the horizon is minutes, the displayed probabilities can swing quickly as price moves and as time remaining collapses. ⁹

From an options perspective, these are effectively *very short-horizon digitals* on whether the return over the window is nonnegative (with "strike" anchored at the opening price).

Trading venue mechanics that matter for market makers

Polymarket's CLOB has several microstructure properties that directly affect 15-minute market making:

- **Displayed probability vs real tradable liquidity:** the "midpoint" is the average of best bid/ask and is displayed as the implied probability, but if the bid-ask spread is wider than **\$0.10**, Polymarket displays the **last traded price** instead of the midpoint—this can distort naive "mid" models if you don't use best bid/ask directly. ¹⁰
- **Tick size regime changes:** the market WebSocket emits `tick_size_change` events "when price hits >0.96 or <0.04," which is "critical for trading bots" because orders using the old tick size will be rejected. This is especially relevant in 15-minute markets because prices often run toward extremes as the window nears expiry. ¹¹
- **Order types and maker controls:** all orders are expressed as limit orders; market orders are implemented by submitting a marketable limit, with support for GTC/GTD and FOK/FAK styles plus **post-only** to guarantee maker behavior, and **batch order** submission up to 15 orders per request. ¹²
- **Heartbeat requirement:** if a valid heartbeat is not received within ~10 seconds, "all open orders are cancelled." For an always-on market maker, heartbeat reliability becomes a first-class operational risk. ¹²
- **Matching engine restarts:** Polymarket publishes a weekly matching engine restart schedule (Tuesdays 7:00 AM ET, ~90 seconds), during which the CLOB API returns HTTP `425` and clients should retry with backoff. This can create sudden liquidity gaps if your bot doesn't handle it correctly. ¹³

Fee and rebate structure as a “market structure edge”

For Crypto-category fee-enabled markets, Polymarket’s fee model is not “flat bps.” It is a convex function of price uncertainty:

- Fee formula: $\text{fee} = C \times \text{feeRate} \times p \times (1 - p)$ where C is shares and p is share price.²
- **Taker-only:** makers are never charged fees; only takers pay them.¹⁴
- Crypto parameters: taker fee rate **0.072**, maker fee rate 0, maker rebate share **20%**.²
- Magnitude at midpoint (illustrative using Polymarket’s own table): for 100 shares at \$0.50, Crypto taker fee is **\$1.80**.¹⁴

Maker rebates are distributed daily in USDC and are computed using a fee-equivalent metric:

- For each executed maker fill, $\text{fee_equivalent} = C \times \text{feeRate} \times p \times (1 - p)$.¹⁵
- Per market, your daily rebate is $(\text{your_fee_equivalent} / \text{total_fee_equivalent}) \times \text{rebate_pool}$, meaning you only compete against other makers in that market.¹⁵

Implication: any bot evaluation that treats gross spread capture as the only return is incomplete. In active 15-minute crypto markets where takers pay meaningful fees near the midpoint, rebates can be a primary P&L line item.²

APIs, smart contracts, and on-chain mechanics for market making

Official API surface and rate limits

Polymarket exposes multiple API families with distinct purposes and rate limits:

- **CLOB API** (trading + live orderbook): base URL `https://clob.polymarket.com` with high burst allowances for many endpoints in its published rate limits (e.g., `/book`, `/price`, `/prices-history`), and separate burst/sustained limits for trading endpoints like `POST /order`.¹⁶
- **Gamma API** (market/event metadata): base URL `https://gamma-api.polymarket.com`, including endpoints like `/events/slug/{slug}` returning event fields including dates, flags (active/closed), and embedded markets.¹⁷
- **Data API** (ledger/history): base URL `https://data-api.polymarket.com` is explicitly called out in the rate limit documentation and includes endpoints such as `/trades`, `/positions`, and `/closed-positions`.¹⁸
- **Relayer API** (gasless transactions for contract wallets): e.g., `POST https://relayer-v2.polymarket.com/submit`, authenticated using Builder or Relayer API keys, supporting transaction `type` values such as `SAFE` and `PROXY`.¹⁹

Rate limiting is enforced via Cloudflare throttling, where you are “throttled (delayed/queued)” rather than immediately rejected when exceeding limits, and windows are sliding; this matters for cancel/replace loops at high frequency.¹⁸

WebSocket real-time data for market making

For market-making, you generally should not poll REST. Polymarket provides a market WebSocket channel:

```
wss://ws-subscriptions-clob.polymarket.com/ws/market
```

The documentation describes subscribing by asset IDs (token IDs) and receiving events including `book` snapshots, `price_change`, `last_trade_price`, and `tick_size_change`; optional events like `best_bid_ask`, `new_market`, and `market_resolved` require `custom_feature_enabled: true`.

11

Wallet and funding mechanics for active market makers

Polymarket's "Getting Started" guide for market makers highlights the practical setup path:

- Market makers need **USDC.e on Polygon** (as currently documented) and can deposit via a Bridge API. ²⁰
- Two main wallet modes are described: an EOA (you pay gas for on-chain actions) or a **Safe wallet** deployed via Polymarket's relay for **gasless transactions** and batching. ²¹
- Required approvals are explicitly listed (USDC.e → CTF contract to split; CTF outcome tokens → exchange contracts to trade). ²⁰
- API credentials: L2 API keys can be derived from your wallet through the CLOB client method `createOrDeriveApiKey()`. ²²

Core on-chain primitives

Polymarket documents that every market position is an ERC-1155 outcome token in the **Gnosis Conditional Token Framework**:

- Each binary market has exactly two tokens (Yes/No) that each redeem for **\$1** if they are the winning token after resolution. ²³
- Tokens are always fully collateralized: each Yes/No pair is backed by **\$1.00 USDC.e** locked in the CTF contract. ²³
- The operational flow for market makers is: **split** (mint inventory), **trade** (CLOB), optionally **merge** (convert matched pairs back to collateral), and **redeem** after resolution. ²³

Polymarket's contract address registry (Polygon, chain ID 137) enumerates the key deployed contracts:

- **CTF Exchange** (standard matching/settlement) and **Conditional Tokens (CTF)** contract address. ²⁴
- **Neg Risk CTF Exchange** and adapter for multi-outcome "negative risk" markets (not the core focus of 15-minute Up/Down, but relevant if your bot generalizes). ²⁵
- **UMA Adapter and UMA Optimistic Oracle** addresses for resolution infrastructure (Polymarket's general resolution process uses UMA optimistic oracle). ²⁶

Exchange protocol design and order structure

Polymarket's open-source **CTF Exchange** repo describes the protocol as an exchange facilitating atomic swaps between CTF ERC-1155 assets and an ERC-20 collateral asset, designed for a “hybrid-decentralized” model: an operator provides off-chain matching, while settlement happens on-chain non-custodially. ²⁷

The exchange documentation details that:

- Orders are EIP-712 typed structured data and include fields such as `maker`, `signer`, `tokenId`, amounts, expiration, nonce, and `feeRateBps` plus a `signatureType`. ²⁸
- Signature types include EOA, Polymarket proxy, Polymarket Gnosis Safe, and EIP-1271 contract signatures. ²⁸

Fee handling for API users

Polymarket's **platform-level** fee documentation states:

- Fee-enabled markets expose `feesEnabled: true` in the market object, and you can query by token ID with `GET /fee-rate?token_id=...`. ²
- The official clients automatically fetch fee rate and include `feeRateBps` in the signed order payload; if you sign manually, you must include `feeRateBps` as part of the signed object and you must fetch dynamically rather than hardcode. ¹⁴

On-chain, Polymarket also publishes an “Exchange Fee Module” contract which proxies the exchange `matchOrders` and can refund fees above an operator's intent; its exposed function includes explicit fee amount parameters (`takerFeeAmount`, `makerFeeAmounts`) in addition to fill amounts. ²⁹

This is a key clue that “fee economics” may involve both signed fee rate parameters and operator-set fee amounts at settlement time (exact behavior depends on your execution path and the active exchange version). ³⁰

Geographic restrictions

Polymarket explicitly restricts order placement based on IP geolocation and publishes a geoblock endpoint on the main site:

```
GET https://polymarket.com/api/geoblock
```

The documentation states that orders from blocked regions are rejected, provides a list of blocked countries and some “close-only” jurisdictions, and recommends builders implement checks before attempting to trade. ³¹

This is a hard constraint for any real-world bot deployment and must be treated as a compliance requirement, not an “optimization problem.” ³¹

Market maker bot architectures and algorithms for 15-minute binaries

Reference architecture

A common high-performance architecture decomposes into: real-time market data, a pricing engine, an inventory/risk engine, an order management system (OMS) with cancel/replace, and accounting/monitoring.

```
flowchart LR
    subgraph Data
        PMWS[Polymarket WebSocket\n(book, best_bid_ask, tick_size_change)]
        PMREST[Polymarket REST\n(book/price/fee-rate/markets)]
        EXT[External price feeds\n(Binance/Coinbase/etc.)]
    end

    subgraph Core
        F[Fair value model\nP(Up) / P(Down)]
        INV[Inventory & risk\nlimits, skew, hedges]
        QUOTE[Quoting engine\nlevels, spreads, sizes]
        OMS[OMS\npost-only, GTD, batch,\nheartbeat, cancel/replace]
    end

    subgraph Ops
        LEDGER[Positions & PnL\nfills, inventory, rebates]
        OBS[Monitoring\nlatency, rejects, restarts]
    end

    PMWS --> F
    PMREST --> F
    EXT --> F

    F --> INV --> QUOTE --> OMS
    OMS --> LEDGER
    LEDGER --> INV
    OMS --> OBS
    PMWS --> OBS
```

This aligns with community bot designs that explicitly combine external exchange feeds and Polymarket execution, and production-grade setups that emphasize monitoring and recovery (e.g., websocket reconnection, rate limiting, simulation/live toggle). ³²

Pricing models for an “Up/Down over 15 minutes” contract

Let S_0 be the “Price to Beat” at window start and S_t be the current price at time t (within the 15-minute interval); the contract resolves “Up” if $S_T \geq S_0$ at the window end T . Polymarket pages define this rule and identify Chainlink data streams as the price source. ⁸

A baseline quantitative model uses a short-horizon diffusion for log-price:

- $\log(S_T) \mid \log(S_t) \sim \text{Normal}(\log(S_t) + (\mu - \frac{1}{2}\sigma^2)\tau, \sigma^2\tau)$, where $\tau = T - t$.
- Then
$$P(\text{Up}) = P(\log(S_T) \geq \log(S_0) \mid \log(S_t)) = 1 - \Phi\left(\frac{\log(S_0) - \log(S_t) - (\mu - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}\right).$$

Practical calibration: σ is typically estimated from very recent realized volatility (seconds-to-minutes) or from an external venue’s short-horizon implied/realized metrics. In 15-minute markets, σ is time-varying and microstructure noise matters; using volatility over too long a window can systematically misprice near the end.

Market making quote formation

Most bots implement one of these approaches:

- **Midpoint $\pm$ spread quoting with inventory skew:** choose a fair probability p^* and place bids/asks around it, skewing toward flattening inventory. This is the “classical” electronic market-making approach described in Avellaneda–Stoikov style models. ³³
- **Avellaneda–Stoikov-style optimal quoting:** compute a reservation price that shifts with inventory and risk-aversion, and a spread that scales with volatility and risk appetite. The original framework studies optimal bid/ask submission under inventory risk in a limit order book. ³³
- **Event-time adaptive quoting:** because the “digital” payoff has extreme convexity as $\tau \rightarrow 0$, many strategies widen spreads or reduce size mechanically as the window approaches the end, regardless of p^* , to control tail outcomes.

Inventory and risk management for 15-minute binaries

Key constraints are mechanical:

- You cannot ignore inventory: outcome tokens are ERC-1155 positions; position value is balance $\times$ current price, and P&L depends on resolution vs entry price. ³⁴
- Market makers often create inventory by **splitting** collateral into both sides (Yes+No pairs), enabling continuous two-sided quoting and pairwise merging as inventory accumulates. ²³

Key risks are statistical:

- **Adverse selection:** in short-horizon price markets, informed flow arrives when the underlying price moves rapidly relative to slower-moving Polymarket quotes (or when the oracle feed update schedule creates “basis risk”).
- **Tail risk (last seconds):** the contract resolves at a single time boundary; a bot that sells “almost certain” outcomes repeatedly can show long runs of small wins and occasional outsized losses. This

is a common pattern in short-dated digital-selling strategies (and it is exactly the type of risk that pure equity-curve screenshots can hide).

Order placement mechanics that matter operationally

A Polymarket market maker must build around the platform's hard operational constraints:

- **Post-only** for maker guarantee (avoid accidental taker fees). ³⁵
- **Tick sizes** and tick-size changes (auto-adjust when `tick_size_change` arrives). ³⁶
- **Fee-rate-aware signing**: you must dynamically fetch and include `feeRateBps` in the signed order; the official clients do this automatically, but custom signers must do it explicitly. ¹⁴
- **Heartbeat**: if you fail heartbeat, orders are cancelled; your system must treat connectivity as risk. ¹²
- **Restart handling**: treat HTTP `425` as transient engine restart and apply backoff. ¹³
- **Balance/allowance validity checks**: orders are monitored; the documentation explicitly warns that makers "caught intentionally abusing these checks will be blacklisted," which means your simulator must model reserved balances correctly and your production system must avoid placing orders it cannot honor. ³⁷

Strategy comparison table

Strategy family	Core edge hypothesis	Typical holding time	Required infra	Primary risk modes	When it tends to work
Passive two-sided quoting near <code>p*</code>	Spread + maker rebates dominate; little directional view	Seconds-minutes; flatten before expiry	WebSocket book + OMS + heartbeat + tick-size handling	Adverse selection on fast moves; inventory drift; sudden tick-size regime shift	High taker flow around mid; stable connectivity; moderate volatility ³⁸
Inventory-skewed quoting (A-S style)	Control inventory risk while staying competitive	Seconds-minutes	All above + solid inventory accounting	Model risk (σ misestimate); poor skew can "lean into" adverse selection	When you must quote continuously without building large directional exposure ³⁹
Latency-informed maker (reactive quoting)	Quote adjusts faster than peers to reduce toxicity	Sub-second-seconds per update	Low-latency WS pipeline; fast cancel/replace; careful rate limit handling	Overtrading; throttle/lag; selection by faster bots	When oracle/external feed moves are frequent and many takers are reactive ⁴⁰

Strategy family	Core edge hypothesis	Typical holding time	Required infra	Primary risk modes	When it tends to work
End-of-window “reduce/withdraw”	Avoid last-seconds jump risk	Final 10–60 seconds	Reliable time sync + OMS + risk rules	Giving up rebates/volume; missing profitable late flow	In assets with high jump risk (news spikes) or thin liquidity near close ⁴¹

Strategy variants across BTC, ETH, SOL, and XRP and why performance differs

Liquidity and flow differences are structural

Polymarket’s 15-minute crypto page lists a small set of live markets per asset, but individual market pages show widely varying activity and sometimes extremely low volume (even \$0 on some windows), implying that *fill opportunity* is highly window-dependent. ⁴²

Since maker rebates are funded by taker fees and distributed per market to makers competing in that market, low-flow windows may generate negligible rebate pools and make “rebate farming” ineffective. ³

Oracle basis risk differs by asset feed behavior

Each of the Up/Down 15-minute markets explicitly points to a Chainlink data stream for the underlying (e.g., ETH/USD, SOL/USD). ⁴³

In practice, any mismatch between (a) the data stream’s update timing and (b) external venue prices creates a “basis” between what the contract resolves on and what many traders watch. This can affect:

- **Toxicity:** if takers trade based on faster external information, makers face adverse selection.
- **Hedge quality:** if you hedge using spot/perps that track a different composite than the oracle stream, you inherit basis risk.

Polymarket’s pages explicitly warn (e.g., on Solana pages) that the market is about price “according to Chainlink data stream ... not according to other sources.” ⁴⁴

Volatility and jump behavior reshape optimal quoting

While exact realized volatility parameters are not provided in Polymarket docs, the *mechanics* of 15-minute digitals imply that higher short-horizon volatility increases:

- The speed at which p^* moves (more cancel/replace pressure).
- Adverse selection costs (more chance your resting quotes are hit right before a move).
- Tail risk near expiry (more chance of last-second reversals).

This tends to push makers toward **wider spreads, smaller sizes**, and **earlier de-risking** in assets with more violent micro-moves, and toward more competitive quoting in deeper, steadier assets (often BTC/ETH relative to smaller alts).

Time-of-day and window effects

Because each market is a discrete 15-minute window, behavior can be highly non-stationary:

- Some windows attract concentrated attention and volume; others are deserts. ⁴⁵
- The “Price to Beat” resets each window, meaning the barrier location changes discretely, and early-window probability is often near 50% absent movement—precisely where fee-equivalent weighting and taker fees are highest. ⁴⁶

Why many bots show linear rising equity curves and how to test those claims

The mechanical P&L drivers that can generate “smooth” equity

Many public bot reports—especially in short-duration markets—show smooth, almost linear equity growth. In Polymarket 15-minute crypto markets, the platform’s design provides at least three structural sources of smoothness:

Maker rebates are paid daily in USDC and are proportional to executed maker flow. The maker rebate program explicitly pays daily rebates, funded by taker fees, with the fee-equivalent weighting

$C \times \text{feeRate} \times p \times (1 - p)$ and Crypto rebate share of 20%. ³

If your bot’s executed maker volume is stable over time (especially near midpoints), the rebate accrual can be near-linear in time.

Taker fee curve concentrates value near $p \approx 0.5$. Polymarket’s own tables show the Crypto taker fee per 100 shares peaks at \$1.80 at $p=0.50$ and declines symmetrically toward extremes. ¹⁴

Because rebate distribution is based on the same fee-equivalent formula, fills near midpoints are disproportionately valuable for rebate share. ¹⁵

Holding rewards can add an additional steady yield in eligible markets: Polymarket documents a 4.00% annualized “Holding Reward” based on total position value, with hourly sampling and daily distribution (variable and discretionary). ³⁴

For 15-minute strategies this is typically secondary, but it can still matter for bots that carry inventory across hours.

Survivorship bias and “short gamma” equity illusions

Even if a strategy is fundamentally risky, it may *appear* safe in limited samples:

- **Short digital risk:** repeatedly selling outcomes priced near 0.95/0.05 can generate many small wins (appearing linear) until a rare reversal causes a large drawdown.

- **Survivorship/selection:** bots that blow up stop posting charts; the remaining charts are biased upward.
- **Accounting artifacts:** equity curves sometimes include rebates as realized cash inflows while marking inventory optimistically (or not marking-to-market at all).

Polymarket itself highlights the importance of correct P&L interpretation: position value is token balance $\times$ current price and you can sell before resolution; P&L depends on resolution vs entry price. ³⁴

Data sources for an empirical investigation

Polymarket provides enough official interfaces to do rigorous measurement, provided you collect the right data:

- **Fee-rate and fee-enabled flags:** `feesEnabled` on market objects and `GET /fee-rate?token_id=...` for market fee rate bps. ¹⁴
- **Trades and positions history:** Data API endpoints are documented and rate-limited; profile endpoints include a “Download an accounting snapshot (ZIP of CSVs)” option in the API reference navigation. ⁴⁷
- **Real-time and snapshot orderbook states:** `GET /book` and the market WebSocket for continuous capture. ¹¹
- **On-chain settlement & token flows:** contract addresses and open-source exchange code allow independent reconstruction from Polygon logs if needed. ⁴⁸

Statistical analyses to explain “linear” returns

If you are evaluating a bot (yours or someone else’s), the minimum rigorous set of diagnostics should include:

- **P&L decomposition** into: spread capture, maker rebates, holding reward, mark-to-market inventory drift, and realized tail events at resolution. Maker rebates explicitly depend on fee-equivalent, so you can compute them per fill using Polymarket’s formula. ³
- **Return distribution** (skew/kurtosis) at multiple horizons: per trade, per 15-minute window, per day. Linear curves often hide fat tails.
- **Adverse selection cost:** measure post-fill price movement vs fill price—particularly around external price jumps.
- **Sensitivity to latency:** measure how fill rate and adverse selection change as your quote update loop slows.

Illustrative charts (conceptual, not from Polymarket data)

```
xychart-beta
title "Illustrative equity curves: smooth accrual vs tail event"
x-axis "Day" [1,2,3,4,5,6,7,8,9,10]
y-axis "Equity (normalized)" 0 --> 1.2
line "Rebates+spread (appears
linear)" [1.00,1.02,1.04,1.06,1.08,1.10,1.12,1.14,1.16,1.18]
```

```
line "Short-digital profile (rare  
drawdown)" [1.00,1.02,1.05,1.07,1.10,1.12,1.15,1.17,0.95,0.98]
```

```
pie title "Illustrative P&L decomposition for a maker bot"  
  "Maker rebates (fee-equivalent share)" : 40  
  "Spread capture" : 35  
  "Directional drift / alpha" : 10  
  "Holding rewards" : 2  
  "Adverse selection + slippage" : 13
```

```
xychart-beta  
  title "Illustrative latency vs fill rate tradeoff"  
  x-axis "Cancel/replace loop latency (ms)" [20,50,100,200,400,800]  
  y-axis "Fill rate (relative)" 0 --> 1  
  line "Fill rate" [0.25,0.40,0.55,0.70,0.80,0.88]  
  line "Adverse selection risk (relative)" [0.90,0.80,0.70,0.55,0.45,0.35]
```

These plots represent patterns you should specifically test for in real data: steady rebate accrual vs structural tail risk in digitals, and fill-rate gains vs toxicity as latency increases.

Backtesting methodology and key parameters to reproduce results

Core backtest challenge: fills, not forecasts

For market making, you cannot validate performance using only midpoint price histories; you must simulate whether your orders would have been filled and at what price, under realistic queue position and latency assumptions.

Polymarket offers:

- Snapshot and real-time orderbook access (REST + WebSocket). ¹¹
- Price history endpoint for tokens (`/prices-history`) suitable for coarse analysis but not sufficient for order-level backtesting in isolation. ¹⁰

If historical full-depth orderbook data is not available for your target period, you must either:

- run a forward “paper market maker” capturing the book in real time, or
- attempt reconstruction from on-chain settlement + off-chain trade records (more complex, and not guaranteed to reproduce queue dynamics).

Recommended backtest design

Use a two-layer simulator:

Event-driven market simulator

- Ingest Polymarket WebSocket events (`book` , `price_change` , `last_trade_price` , `tick_size_change`) to reconstruct the evolving book for each token. ¹¹
- Apply deterministic bot logic: compute fair `p*` , generate quotes at valid tick sizes, enforce order size constraints, and issue cancel/replace consistent with your loop latency and rate limits. ⁴⁹

Fill and queue model

- Baseline: “top-of-book fill” assumption (optimistic) if your order is at best bid/ask.
- More realistic: probabilistic queue model with partial fills, where fill probability depends on (a) your price aggressiveness, (b) your time at level (latency), and (c) observed trade-through volume.

Key parameters you must specify (and publish) to make results reproducible

Operational parameters

- Quote refresh interval (ms) and distribution (jitter).
- Cancel/replace latency (MS round-trip) and failure handling.
- Heartbeat interval and drop behavior (orders cancelled if heartbeat not maintained). ¹²
- Engine restart handling for HTTP 425 windows. ¹³
- Tick-size change handling logic. ¹⁰

Market/contract parameters

- Fee rate (`feeRateBps`) and `feesEnabled` — fetched dynamically; never hard-coded. ¹⁴
- Maker rebate share and daily pool allocation modeling (per market). ¹⁵
- Resolution rule: Up iff $\text{end} \geq \text{start}$, using Chainlink stream (affects any “hold-to-expiry” inventory). ⁴³

Risk parameters

- Max inventory per token/window.
- Time-to-expiry de-risk schedule (e.g., stop quoting in final N seconds).
- Max notional per quote level and per market.
- Optional hedge rules (if you hedge externally).

Evaluation metrics

- Daily and per-window P&L with decomposition (rebates vs trading). ¹⁵
- Max drawdown, tail loss quantiles, and “time to recovery.”
- Inventory distribution at window end.

Implementation, latency, security, MEV, and regulatory considerations

Practical infrastructure tiers

Tier	What you can realistically do	Required components	Typical failure modes
Prototype	Paper making, slow refresh, basic analytics	Official client (TS/Py/Rust), REST polling, simple DB	Missed tick-size changes; heartbeat/order cancellations; stale quotes ⁵⁰
Production basic	Continuous quoting in a few markets	WebSocket market channel, OMS with post-only & batch, robust heartbeat & retry, monitoring	Throttling under burst cancels; restart gaps; inventory drift ⁵¹
Low-latency	Reactive maker, many markets/windows	VPS near API edge, high-perf event loop, in-memory book, persistent state store, strict time sync	Over-canceling, rate-limit bottlenecks, data feed desync, hidden tail losses
Institutional-grade	Scale + auditability + key safety	Safe wallet flows, relayer integration, remote signing, full accounting snapshots, alerting	Key compromise risk; operational misconfig; compliance failures ⁵²

On-chain and MEV/front-running risks

Even though Polymarket uses off-chain matching plus on-chain settlement, any system that settles via smart contracts inherits **transaction ordering and MEV** risks in principle. “Flash Boys 2.0” documents widespread bot-driven exploitation and “miner/extractor value” in decentralized trading contexts due to transaction reordering and frontrunning dynamics. ⁵³

For a Polymarket market-maker bot, this translates into practical security/robustness advice:

- Prefer **post-only** to avoid accidentally becoming taker and paying high mid-curve fees. ³⁵
- Use Safe/relayer flows where appropriate and avoid hot private keys on a public server if you can (Polymarket supports Safe wallets deployed via relayer). ²¹
- Maintain strict observability around any on-chain transaction submission delays (especially if you do splits/merges frequently). ⁵⁴

Security checklist for bot operators

Minimize irreversible loss modes:

- Use least-privilege approvals rather than unlimited where feasible (Polymarket docs show MaxUint approvals as an example; treat as a tradeoff). ⁵⁵

- Isolate signing keys; consider contract wallets + relayer. ²¹
- Implement “kill switches”: if heartbeat fails, if engine returns 425 repeatedly, if WS book lags, flatten inventory and stop quoting. ⁵⁶

Regulatory and ethical considerations

- **Geographic restrictions:** Polymarket explicitly blocks order placement from certain jurisdictions and instructs builders to check the geoblock endpoint; bypassing or evading such restrictions is not just unethical but likely a ToS/compliance violation. ³¹
- **Market integrity:** strategies intended to manipulate odds, spoof liquidity, or exploit oracle mechanics raise major ethical issues and may violate platform rules; Polymarket explicitly monitors orders for validity and warns of blacklisting for abuse. ¹²
- **Separate regulated venue context:** Polymarket US materials (distinct from polymarket.com) describe a regulated fee schedule and rebate coefficients using the same $0 \times C \times p \times (1-p)$ structure; if you are operating in or targeting US-regulated infrastructure, you must use the correct venue’s rules and APIs. ⁵⁷

Actionable recommendations and next-step checklist

Recommendations for building or evaluating a 15-minute crypto market maker

Build around the fee/rebate reality first:

- Treat maker rebates as a core revenue line: model $\text{fee_equivalent} = C \times \text{feeRate} \times p \times (1 - p)$ per fill and estimate rebate share; don’t evaluate performance on spread capture alone. ³
- Focus quoting near midpoints only if you can manage toxicity; midpoints maximize fee-equivalent weight but also attract informed flow during volatile moves. ²

Engineer for platform-specific failure modes:

- Implement WebSocket-first book ingestion and explicitly handle `tick_size_change`. ¹¹
- Treat heartbeat as mandatory infrastructure, not “nice to have.” ¹²
- Implement restart logic for weekly engine restarts (HTTP 425). ¹³
- Never hardcode fee parameters; fetch `feeRateBps` dynamically and sign it into the order. ¹⁴

Control tail risk explicitly:

- Add “time-to-expiry risk ramps”: reduce size and/or widen spreads as τ shrinks; a digital’s tail risk is concentrated at the boundary.
- Run stress tests where price reverses sharply in the last 10–30 seconds and measure worst-case drawdown at your chosen size limits.

Validate with fill-realistic backtests:

- If you don’t have historical depth-of-book, run forward data capture with WebSockets and paper trading to build a ground-truth dataset. ⁵⁸

- Decompose P&L into rebates vs trading edge; many “linear” curves are simply high-volume rebate harvesting plus mild spread capture. ³

Concise checklist of next steps

Operational setup

- Confirm legal eligibility using the geoblock endpoint and your deployment IP region. ³¹
- Obtain a Polymarket wallet setup (EOA or Safe) and complete approvals for USDC.e→CTF and CTF→Exchange. ²⁰
- Derive CLOB API credentials via the official client. ²²

Market connectivity

- Implement WebSocket market subscriptions for token IDs and handle `book`, `best_bid_ask`, and `tick_size_change`. ¹¹
- Implement heartbeat polling and automatic recovery (WS reconnect, REST retry/backoff, engine 425). ⁵⁶

Strategy and risk

- Implement a baseline fair-probability model and inventory-skew quoting; start conservative with sizes and widen spreads near expiry.
- Implement P&L decomposition: spread, rebates, holding reward (if applicable), and adverse selection. ⁵⁹

Backtesting and evaluation

- Capture real-time orderbook/trade streams for several weeks across BTC/ETH/SOL/XRP windows; build an event-driven simulator with latency and queue assumptions. ⁶⁰
- Report results with: max drawdown, tail quantiles, and scenario stress tests—not just a single equity curve.

Security and governance

- Treat key custody as a first-order design choice; consider Safe + relayer flows where appropriate. ²¹
- Add a hard “kill switch” for anomaly detection (fee-rate change, tick-size change mismatch, repeated rejects, WS lag).

If any specific dataset you need (e.g., historical full-depth orderbooks for backtesting fills) is not available via Polymarket’s APIs, that should be stated explicitly in your methodology and substituted with forward data capture via WebSockets, since fill realism is the core determinant of whether a 15-minute market maker is genuinely profitable. ⁶¹

¹ ⁸ ⁹ ⁴³ ⁴⁶ <https://polymarket.com/event/eth-updown-15m-1775855700>

<https://polymarket.com/event/eth-updown-15m-1775855700>

2 14 <https://docs.polymarket.com/trading/fees>
<https://docs.polymarket.com/trading/fees>

3 15 38 59 <https://docs.polymarket.com/market-makers/maker-rebates>
<https://docs.polymarket.com/market-makers/maker-rebates>

4 23 34 <https://docs.polymarket.com/concepts/positions-tokens>
<https://docs.polymarket.com/concepts/positions-tokens>

5 24 25 26 48 <https://docs.polymarket.com/resources/contract-addresses>
<https://docs.polymarket.com/resources/contract-addresses>

6 10 11 36 40 41 51 58 60 61 <https://docs.polymarket.com/trading/orderbook>
<https://docs.polymarket.com/trading/orderbook>

7 42 <https://polymarket.com/crypto/15M>
<https://polymarket.com/crypto/15M>

12 35 37 49 56 <https://docs.polymarket.com/trading/orders/create>
<https://docs.polymarket.com/trading/orders/create>

13 <https://docs.polymarket.com/trading/matching-engine>
<https://docs.polymarket.com/trading/matching-engine>

16 18 47 <https://docs.polymarket.com/api-reference/rate-limits>
<https://docs.polymarket.com/api-reference/rate-limits>

17 <https://docs.polymarket.com/api-reference/events/get-event-by-slug>
<https://docs.polymarket.com/api-reference/events/get-event-by-slug>

19 <https://docs.polymarket.com/api-reference/relayer>
<https://docs.polymarket.com/api-reference/relayer>

20 21 22 52 55 <https://docs.polymarket.com/market-makers/getting-started>
<https://docs.polymarket.com/market-makers/getting-started>

27 <https://github.com/Polymarket/ctf-exchange>
<https://github.com/Polymarket/ctf-exchange>

28 <https://mintlify.wiki/Polymarket/ctf-exchange/concepts/order-structure>
<https://mintlify.wiki/Polymarket/ctf-exchange/concepts/order-structure>

29 30 <https://github.com/Polymarket/exchange-fee-module>
<https://github.com/Polymarket/exchange-fee-module>

31 <https://docs.polymarket.com/api-reference/geoblock>
<https://docs.polymarket.com/api-reference/geoblock>

32 <https://github.com/aulekator/Polymarket-BTC-15-Minute-Trading-Bot>
<https://github.com/aulekator/Polymarket-BTC-15-Minute-Trading-Bot>

33 39 <https://math.nyu.edu/inmemoriam/avellaneda/HighFrequencyTrading.pdf>
<https://math.nyu.edu/inmemoriam/avellaneda/HighFrequencyTrading.pdf>

44 <https://polymarket.com/event/sol-updown-15m-1775639700>
<https://polymarket.com/event/sol-updown-15m-1775639700>

45 <https://polymarket.com/event/sol-updown-15m-1775385000>

<https://polymarket.com/event/sol-updown-15m-1775385000>

50 <https://docs.polymarket.com/api-reference/clients-sdks>

<https://docs.polymarket.com/api-reference/clients-sdks>

53 <https://arxiv.org/abs/1904.05234>

<https://arxiv.org/abs/1904.05234>

54 <https://docs.polymarket.com/trading/ctf/overview>

<https://docs.polymarket.com/trading/ctf/overview>

57 <https://docs.polymarket.us/fees>

<https://docs.polymarket.us/fees>